The Honorable Laura V. Swett, Chairman
The Honorable David Rosner, Commissioner
The Honorable David A. LaCerte, Commissioner
The Honorable Lindsay S. See, Commissioner
The Honorable Judy W. Chang, Commissioner

Federal Energy Regulatory Commission
888 First Street, NE
Washington, DC 20426

Re: Responding to the Advanced Notice of Proposed Rulemaking Regarding the Interconnection of Large Loads to the Interstate Transmission Systems (Docket No. RM26-4-000)

Dear Chairman Swett,

Helion Energy appreciates the opportunity to provide comments in response to the Advanced Notice of Proposed Rulemaking (ANOPR) submitted to the Federal Energy Regulatory Commission (FERC) by Secretary Wright.[1] As a U.S.-based fusion energy company delivering the world's first commercial fusion power plant by 2028 with plans to rapidly scale deployment thereafter, Helion is deeply invested in ensuring that federal policy supports accelerated interconnection of baseload and dispatchable generation that can meet America's surging electricity demand.[2]

The United States is entering an era of explosive electricity demand driven by artificial intelligence, advanced manufacturing, and industrial reshoring.[3] At the same time, the processes for adding new generation and transmission are too slow and cumbersome to meet the demands of America's competitive environment. Regional operators are warning of capacity shortfalls as early as this year,[4] with projects delayed years by interconnection queues.[5] Developers are forced to choose where to build based not on where power is needed most, but on where approvals can be secured fastest. More than half of U.S. developers now identify interconnection delays as the single biggest barrier to adding new generation capacity.[6] That is why Helion believes it is essential to simultaneously address the ANOPR comments and to reevaluate and improve generation interconnection practices in a single rulemaking process that prioritizes an interconnection process capable of meeting the moment.

To address these challenges, Helion recommends FERC:

1. Prioritize Baseload, Dispatchable Resources in Interconnection Processes
2. Streamline Interconnection for Collocated Generation While Preserving Grid Reliability
3. Leverage Technology for Rapid Interconnection Study and Approval
4. Integrate Advanced Technologies into Long-Term Planning

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[1] Secretary Wright to Chairman Rosner and Commissioners, "*Secretary of Energy's Direction that the Federal Energy Regulatory Commission Initiate Rulemaking Procedures and Proposal Regarding the Interconnection of Large Loads Pursuant to the Secretary's Authority Under Section 403 of the Department of Energy Organization Act*," October 2025.
[2] Helion Energy, "*Helion Secures Land and Begins Building on the Site of World's First Fusion Power Plant*," July 2025.
[3] U.S. Energy Information Administration, "*After more than a decade of little change, U.S. electricity consumption is rising again*," May 2025.
[4] North American Electric Reliability Corporation, "*Rising Demand, Evolving Resources Continue to Challenge Winter Grid Reliability*," November 2025.
[5] Berkeley Lab, "*Queued Up: Characteristics of Power Plants Seeking Transmission Interconnection*," April 2024.
[6] Third Way, "*Picking Up the PACE: A Comprehensive Analysis of Available Pathways to Accelerating Clean Energy (PACE)*," November 2025.

5. Establish Clear Pathways for Emerging Technologies

The Race to Build: Why Interconnection Reform Determines U.S. Leadership

As electricity demand surges, the speed at which we can connect new loads and generation to the grid will determine whether the nation leads or lags in the global competition for economic prosperity and energy security.

Advanced energy technologies—whether nuclear, fusion, geothermal, or grid-scale storage—cannot scale at the speed of innovation if they must wait for years in queues designed for another era. Data centers powering AI development cannot drive American technological leadership if they're built in other countries with faster grid access. Manufacturing facilities cannot break ground if reliable power takes half a decade to secure.

China is moving at a pace that should serve as a wake-up call.[7] It adds gigawatts of new capacity in the time it takes U.S. projects to complete initial studies by aligning industrial policy, manufacturing, and energy deployment with ruthless efficiency.[8,9] The U.S., by contrast, is struggling to connect projects in a reasonable timeframe—dangerously unprepared for the scaling that our economic and national security demand.

FERC has a once-in-a-generation opportunity to break the logjam that has made our own regulatory process the biggest barrier to American progress. Streamlined interconnection pathways will determine whether next-generation technologies are deployed domestically or abroad, and whether American workers or foreign competitors capture the industries and strategic advantages of the 21st-century energy economy.

Modern Grid Challenges and the Case for Baseload, Dispatchable Power

America's future energy system will depend on baseload, dispatchable resources that can operate in direct coordination with large, concentrated loads.

The grid's challenge is not just capacity—it's controllability. As data centers, AI facilities, and advanced manufacturing drive unprecedented demand growth, the system needs generation that can ramp precisely to match real-time load fluctuations.

This mismatch forces operators to rely on spinning reserves, overbuilt capacity, or expensive peaking plants that run inefficiently. FERC's framework should prioritize resources capable of providing baseload, dispatchable, non-fuel-limited power that enhances grid stability. It is time for FERC to recognize that non-discriminatory transmission access does not prevent—and actually supports—treating dispatchable, baseload power with priority to meet community energy demand, support national security concerns, and secure America's competitive edge.

Helion's fusion plants will deliver baseload power that is not fuel-limited with rapid dispatchability—ramping output dynamically in response to grid conditions within seconds through our inherent pulsed power system. This combination of reliability and responsiveness addresses the operational challenges that currently constrain grid expansion.

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[7] Special Competitive Studies Project, "*Cash, Scale, and Speed: Why China's $6.5 Billion Fusion Buildout Should Shock the World*," September 2025.
[8] U.S. Energy Information Administration, "*China's solar capacity installations grew rapidly in 2024*," April 2025.
[9] The Economist, "China's Clean Energy Revolution Will Reshape Markets and Politics," November 2025.

Collocated Generation: Reducing Congestion While Expanding Access

There is a fundamental shift in how America must approach grid planning: the rise of large, concentrated loads requiring direct coordination with generation resources. This insight points toward a solution that reduces transmission congestion while enhancing reliability—collocating generation directly at load centers.

The case for collocation is particularly compelling for large industrial and AI-driven facilities. Calling data centers "flexible" does not mean they can simply absorb sudden cuts in power; real flexibility comes from collocated generation that can ramp up and down with their needs and support the grid in the process. Forcing these customers to curtail operations risks data integrity, and advanced manufacturing cannot simply power down mid-process. The solution is not forcing flexibility where it doesn't exist—it's prioritizing and deploying generation that can match load dynamically, precisely where demand occurs.

Helion's independence from geographic constraints, continuous fuel supply chains, and weather conditions enables siting directly alongside data centers, manufacturing facilities, and other large users. This potential proximity delivers multiple benefits: it reduces stress on the bulk transmission system, eliminates line losses, reinforces local reliability, and allows power to flow precisely where it's needed, when it's needed. When baseload, dispatchable, non-fuel-limited generation is collocated with large loads, both the interconnection challenge and the generation/transmission capacity challenge can be addressed simultaneously.

Policy Recommendations

Based on the challenges and opportunities outlined above, Helion respectfully recommends the following actions:

1. Prioritize Baseload, Dispatchable Resources in Interconnection Processes
   FERC should establish clear criteria that recognize and prioritize baseload, dispatchable resources based on their contribution to grid reliability. This should include:
   - Explicit definitions of "baseload, dispatchable, non-fuel limited" generation in market rules and interconnection agreements
   - Expedited study pathways for resources providing baseload, dispatchable, non-fuel-limited power
   - Queue priority weighted by factors like dispatchability and baseload energy capability that promote grid reliability, not solely application date or readiness
2. Streamline Interconnection for Collocated Generation While Preserving Reliability
   This ANOPR identifies the need to coordinate the interconnection of load and generation together. FERC should:
   - Establish simplified procedures for projects pairing large loads with collocated baseload, dispatchable generation that addresses reliability issues
   - Create expedited approval pathways for behind-the-meter or directly connected generation that reduces demand on shared grid infrastructure
   - Ensure projects reducing strain on the broader grid receive appropriate recognition in cost allocation and study timelines

3. Leverage Technology for Rapid Study and Approval
   FERC should modernize interconnection study processes to match today's technological capabilities:
   - Require transmission providers to deploy AI and advanced modeling tools to automate and accelerate interconnection studies for all new load and generation requests
   - Establish standards for real-time or near-instantaneous study results, particularly for projects with minimal transmission impact or those pairing loads with baseload, dispatchable generation
   - Set maximum study completion timelines that decrease progressively as automated systems are developed and deployed
   - Mandate meaningful transparency to increase accountability for missed study deadlines and establish a backstop FERC-led process to address prolonged delays

4. Integrate Advanced Technologies into Long-Term Planning

   FERC's planning frameworks must account for emerging technologies that fundamentally change siting flexibility and reliability. Specifically:

   - Incorporate fusion energy and any other advanced baseload, dispatchable, non-fuel limited technologies into transmission planning models and resource adequacy assessments
   - Update planning assumptions to reflect that certain future resources can be sited independent of fuel supply, water access, or geographic constraints
   - Ensure market structures recognize and value technologies combining baseload capability with rapid dispatchability and flexible siting

5. Establish Clear Pathways for First-of-a-Kind Technologies
   New technologies critical to U.S. competitiveness should not face insurmountable barriers. FERC should:
   - Create expedited review processes for novel generation technologies demonstrating baseload, dispatchable characteristics
   - Establish technical working groups to develop interconnection standards for emerging technologies before they reach commercial scale
   - Ensure first-mover projects receive prioritized study and approval

The interconnection challenges facing America's energy system are policy choices, not technical limitations. FERC has the authority to remove the regulatory barriers preventing the United States from deploying generation at the speed our economy and national security demand.

This ANOPR represents a critical opportunity. By prioritizing baseload, dispatchable resources, streamlining collocated generation, and accelerating study processes, FERC can ensure that America builds the energy infrastructure to power AI leadership, advanced manufacturing, and industrial competitiveness—before competitors like China establish insurmountable advantages in the global energy economy.

The race to build the energy system of the future will be won by the nation that can connect new baseload, dispatchable generation fastest. We urge FERC to ensure that nation is the United States.